



HAMZA HASSAN

MATERIAL SCIENCE ENGINEER

CONTACT

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Park View Town, Park Road,
Islamabad, Pakistan

EDUCATION

2018 - 2022

INSTITUTE OF SPACE TECHNOLOGY,
ISLAMABAD

- Bachelor's of Science in Material Sciences & Engineering
- CGPA: 2.92/ 4.0

Technical Skills

- Project Management
- Research & Development
- Material Characterization
- Material Testing
- Failure Analysis
- Auto CAD
- Microsoft office
- Technical writing
- RLA
- Material selection
- Quality control
- NDT / DT
- ImageJ

LANGUAGES

- English (Fluent)
- Urdu (Fluent)

PROFILE

Proactive Materials Science Engineer with a strong drive for innovation and practical problem-solving. Experienced in applying advanced material analysis and testing to deliver reliable, data-driven solutions for complex industrial challenges. Thrives in collaborative, field-focused environments that demand precision, creativity, and technical excellence.

WORK EXPERIENCE

Jun 2022 - Mar 2025

DEEP CHAIN SOLUTIONS

Inspection Engineer

- Hands-on experience relevant to mechanical testing of material samples using tensile, creep, fatigue, impact, flexural, and hardness tests according to client specifications.
- Independently performed material requisition, inspection, quality control according to company SOPs.
- Documented inspection findings and prepared standard reports including material testing, RCFA, reverse engineering of OEM parts, and RLA assessments for clients.
- Scanning Electron Microscope (SEM) installation, service and maintenance.

Major projects

1- Remaining Life Assessment of Boiler Tubes (SS 304)

- Performed accelerated creep rupture tests, stress analysis, and mechanical testing to assess superheater tube health in a fertilizer plant, determining operational lifespan and replacement needs.

2- Health Evaluation & failure analysis of HRSG tubes (Carbon Steel)

- Conducted material characterization, fractographic and thermal analyses, identifying creep damage as the primary failure mode.

3- Root cause failure analysis of a Axle shaft (Aviation vehicle)

- Utilized OES, metallography, micro Vickers hardness testing, and SEM to determine root cause of brittle fracture, providing recommendations to prevent recurrence.

4- Material characterization of Skid shoe and Handle assembly (Helicopter)

- Conducted mechanical and microstructural testing and other tests on OEM and locally reverse-engineered parts, determining the superior option for operational use.

5- Review of micrographs of center pipe of boiler tube

- Quantified sigma phase content using ImageJ analysis and provided recommendations on whether to replace the pipe or continue service until the next scheduled shutdown.

6- Lab Inspection & Manpower Assessment

- Evaluated laboratory quality control, documentation, and service history to improve material testing operations. Conducted manpower assessment to document each employee's trade and expertise for optimized workforce utilization.

SKILLS

- Project Management
- Public Relations
- Teamwork
- Time Management
- Leadership
- Effective Communication
- Critical Thinking

Material Science

- Performed inspection and standardized testing of materials to verify quality and compliance.
- Characterized materials through microstructural, chemical, and mechanical analysis.
- Executed material testing methods including hardness testing, tensile testing, and spectroscopy.
- Prototyping of injection molding machine along with its designing and 3D printing for IoT sensor.
- Mechanical assembling of 3D printed parts and resolving issues related to 4th Gen design (IoT sensors).
- Inspection and Testing of Emoldino IoT sensor.
- Designed Heat controlled mechanism for IoT sensor.

Research & Development

- Research and Development on High Temperature Polymers for IoT sensor casing and bracket.
- Energy harvesting materials for a self powered future.
- R&D on Adhesive polymers for application of IoT sensor to Injection molding machine.
- Conducted research on anti-corrosive metallic coatings for evaluating their life span in marine environment.
- Research and development of low cost anti-friction composite coating with good adhesion properties.

Autocad fusion 360

- Designing of injection molding machine prototype, IoT sensor case, high temperature bracket, new sensor design, potential issues solution design.

FINAL YEAR PROJECT

Sep 2021 - Jun 2022

Development of Low Emissivity Infrared Camouflage Paint

- A high temperature resistive paint with low radiation emissivity in Infrared region. (NESCOM FUNDED)

CERTIFICATIONS

CPD Certification

- Webinar on 'Using Tools for Quality and Productivity' 6th August, 2025.
- Webinar on 'Entrepreneurship for engineers, PEC, 20th March, 2025.
- Webinar on 'Industrial Automation and smart manufacturing', PEC, 16th March, 2025.
- Workshop on 'Thermal Spray Coatings', Institute of Space Technology, Islamabad, 13th June, 2024.
- Workshop on 'Occupational Health & Safety (OHS), Institute of Space Technology, Islamabad, 24~26th March, 2021.

TRAININGS/AWARENESS SESSIONS

- Auto desk fusion 360 (Cousera 2023)
- Toyota 5s conversation program (2022)
- PEC Registered Engineer (2022)
- Foster Flagship Training program (2019)
- Organizer and Management certificates (by IST)